Expected or observed?

Year 6 Statistics and Probability · Lesson 4 · AC9M6P02 · Version S

NAMES _____ CLASS _____

YOUR BAG

Your bag holds 4 counters: 2 RED, 1 BLUE, 1 GREEN.
You will draw a counter 20 times, putting it back every single time.

Part A · Before you draw anything

Fill in the probability and expected columns FIRST. Do not draw a counter yet.

COLOUR	PROBABILITY	EXPECTED out of 20	OBSERVED	DIFFERENCE observed – expected
RED	2/4 = 50%			
BLUE	1/4 = 25%			
GREEN	1/4 = 25%			
TOTAL				

HOW TO FIND THE EXPECTED NUMBER

Method 1 — divide, then multiply. 20 ÷ 4 = _____ , then × 2 = _____ for red.
Method 2 — decimal first. 2 ÷ 4 = _____ , then × 20 = _____ for red.
Use a calculator for the dividing if you need one.

Part B · Now draw

Tally every draw. Put the counter back each time.

RED

BLUE

GREEN

Now go back and fill in the OBSERVED and DIFFERENCE columns in Part A.

WORKING SPACE

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Part C · Show your thinking

SF

1. A bag has 8 counters. 4 are red. You draw 40 times.

How many reds do you expect? Show your working.

SF

2. A spinner has 4 equal sections. 1 is shaded. You spin 20 times.

How many times do you expect it to land on shaded?

CF

3. Look at the DIFFERENCE column in Part A. Add the three numbers together.

What do you get? Why does that happen?

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Exit ticket

Hand this in at the end of the lesson.

THE SPINNER

A spinner has 5 equal sections. 2 of them are shaded.
You spin it 30 times.

1. How many times do you expect it to land on shaded?

Show your working.

2. Sam spun it 30 times and it landed on shaded 9 times.

Did Sam make a mistake? Explain.

3. Sam says "the probability of shaded is 9 out of 30".

Is Sam right? Explain your answer.

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